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# Linked Open Data Application for Movie Knowledge Graph

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## Abstract

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## 1 Introduction

The rapid growth of digital media has produced a vast amount of structured information about movies, actors, directors, production companies, genres, and ratings. Platforms such as TMDB and IMDb maintain millions of records, and this data is widely used for search, recommendation, and analytics. However, it is often scattered across isolated databases and tabular datasets, each with its own schema, identifiers, and access method. As a result, it is difficult to query across sources, to connect semantically related entities or to ask questions that span several entity types. Knowledge graphs address this problem by representing entities and their relationships in an interconnected graph structure, supporting integration, exploration, and reasoning across heterogeneous information sources [Hogan et al., 2021]. This perspective is consistent with the Linked Data principles of using dereferenceable identifiers and links to related resources [Berners-Lee, 2006].

This work develops a Linked Open Data (LOD) application centered on the movie domain. It covers movies and related entities, including people, organizations, genres, countries, languages, and ratings, together with relationships such as cast membership, directorship, production affiliation, and publication date. The resulting knowledge graph is designed as a reusable and interoperable semantic resource, while remaining focused on a representative subset of the domain that can be modeled consistently using RDF, OWL, and Schema.org vocabulary [Schema.org, 2025, W3C RDF Working Group, 2014]. Reusing Schema.org terms rather than inventing new ones lowers the barrier for other applications to interpret the data.

We demonstrate a complete pipeline for building a movie knowledge graph in accordance with LOD principles. As illustrated in Figure 1, the proposed workflow consists of six stages: (1) defining an ontology for the movie domain; (2) collecting relevant movie data from TMDB; (3) transforming the collected CSV data into RDF in Turtle notation using stable URIs; (4) building an OWL ontology that defines the domain classes, properties, and restrictions; (5) establishing links to DBpedia and Wikidata to improve interoperability and achieve a higher level of Linked Open Data integration; and (6) providing a SPARQL interface through Apache Jena Fuseki for querying the resulting graph [W3C SPARQL Working Group, 2013]. As an optional stage, we provide a static web application, hosted on GitHub Pages in the style of DBpedia, to support browsing and exploration of the generated data.

This remainder of this report is organized as follows. ...

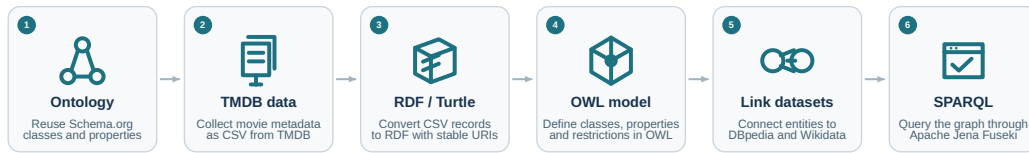


Figure 1: Overview of our method.

## 2 Methodology

## 3 Data collection and transformation

## 4 Ontology constructing and Entity linking

## 5 Evaluation

## 6 Discussion

### Limitations

### Future work

## 7 Conclusion

## References

- Tim Berners-Lee. Linked data. <https://www.w3.org/DesignIssues/LinkedData.html>, 2006. Accessed 2026-10-05.
- Aidan Hogan, Eva Blomqvist, Michael Cochez, Claudia d’Amato, Gerard de Melo, Claudio Gutierrez, Sabrina Kirrane, Jose Emilio Labra Gayo, Roberto Navigli, Sebastian Neumaier, Axel-Cyrille Ngonga Ngomo, Axel Polleres, Sabbir M. Rashid, Anisa Rula, Lukas Schmelzeisen, Juan Sequeda, Steffen Staab, and Antoine Zimmermann. Knowledge graphs. *ACM Computing Surveys*, 54(4), 2021. doi: 10.1145/3447772.
- Schema.org. Getting started with schema.org. <https://schema.org/docs/gs.html>, 2025. Accessed 2026-10-05.
- W3C RDF Working Group. Rdf 1.1 concepts and abstract syntax. <https://www.w3.org/TR/rdf11-concepts/>, 2014. W3C Recommendation; accessed 2026-10-05.
- W3C SPARQL Working Group. Sparql 1.1 query language. <https://www.w3.org/TR/sparql11-query/>, 2013. W3C Recommendation; accessed 2026-10-05.